Math 110 — Week 10

Orthogonality (Notes Ch. 18, LADR 6B–6C, HW10)

§1 Facts used from earlier weeks

V is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Shorthand: "S kills v" means $Sv = 0$, and "q kills T" means $q(T) = 0$. From Week 9: inner products and norms, the orthogonal decomposition $u = cv + w$ with $c = \langle u, v\rangle / \|v\|^2$ and $w \perp v$ (LADR 6.13), Pythagoras, and Cauchy–Schwarz.

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

Why.

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: T of any vector is a combination of the Tu_i (which are 0) and the Tv_j.
3. Independent: if $\sum c_j T v_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the u_i, and independence of the basis of V gives all $c_j = 0$.

(T2) **Dimension shortcuts.**

(a) An independent list of n vectors is a basis.

(b) A subspace of dimension n is all of V.

(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Why.

1. (a), (b): an independent list extends to a basis, and every basis has length n.
2. (c): a basis of U followed by a basis of W is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length n, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis x_i, u_j of U and, separately, to a basis x_i, w_l of W. Together the x_i, u_j, w_l span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ (x, u, w combinations of the x_i, u_j, w_l), then $w = -x - u \in U \cap W$, so w is a combination of the x_i. The basis x_i, w_l of W is independent, so $w = 0$. Then $x + u = 0$, and independence of x_i, u_j gives $x = u = 0$.

(T3) **Products of operators.** If S, T are injective, so is ST. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

Why. $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T4) **Polynomials in T commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are T-invariant.

Why. Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T5) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda)=0 \iff (z-\lambda) \mid p$.

Why. Divide p by $z-\lambda$: the remainder is a constant, and evaluating at λ shows it equals $p(\lambda)$.

(T6) **Eigenvalues.**

(a) λ is an eigenvalue $\iff$ $T-\lambda I$ is not injective.

(b) If $Tv=\lambda v$, then $p(T)v=p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

Why.

1. (a) is the definition: $Tv=\lambda v$ with $v\neq 0$ means $(T-\lambda I)v=0$ with $v\neq 0$.
2. (b): $T^k v=\lambda^k v$ by induction; add up the terms of p.
3. (d): take $v\neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v=0$ for some $p\neq 0$; p is not constant since $v\neq 0$.
4. Factor $p=c(z-\lambda_1)\cdots(z-\lambda_m)$ (T5). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v\neq 0$ to 0, so some $T-\lambda_j I$ is not injective (T3).

(T7) **Upper-triangular matrices.** $\mathcal{M}(T)$ is upper-triangular in $v_1,\ldots,v_n \iff$ every $\operatorname{span}(v_1,\ldots,v_j)$ is invariant. Then the eigenvalues of T are the diagonal entries. Over $\mathbb{C}$ every operator has an upper-triangular matrix.

Why.

1. Column j holds the coordinates of Tv_j, so "upper-triangular" means $Tv_j \in \operatorname{span}(v_1,\ldots,v_j)$ for every j.
2. If $a_{jj}=\lambda$: $T-\lambda I$ maps $\operatorname{span}(v_1,\ldots,v_j)$ into $\operatorname{span}(v_1,\ldots,v_{j-1})$, a smaller space, so it is not injective (T1). So λ is an eigenvalue.
3. If no a_{jj} equals λ: $\mathcal{M}(T-\lambda I)$ is triangular with nonzero diagonal. Its columns are independent: in a relation, the right-most column used is the only one with a nonzero entry in its own diagonal row. So $T-\lambda I$ is injective.
4. Existence over $\mathbb{C}$, by induction on n: take an eigenvalue λ (T6(d)). $U=\operatorname{range}(T-\lambda I)$ is invariant (T4) and $\dim U<n$, since $T-\lambda I$ is not injective (T1).
5. By induction U has a basis $u_1,\ldots,u_m$ with every $\operatorname{span}(u_1,\ldots,u_j)$ invariant. Extend it to a basis $u_1,\ldots,u_m,v_1,\ldots,v_k$ of V.
6. $Tv_j=(T-\lambda I)v_j+\lambda v_j \in U+\operatorname{span}(v_j) \subseteq \operatorname{span}(u_1,\ldots,u_m,v_1,\ldots,v_j)$. So every span of an initial part of the list is invariant.

§2 This week

§2.1 Orthogonality (Ch. 18)

The question. In a general basis, finding coordinates means solving a linear system. If the basis vectors are perpendicular unit vectors, each coordinate is one inner product. We want such bases, a way to build them, and to use them to split spaces into perpendicular pieces.

Definition (Def. 46)**.** $u \perp v$ if $\langle u,v\rangle=0$. $e_1,\ldots,e_m$ is **orthonormal** if $\langle e_i,e_j\rangle=\delta_{ij}$.

Proposition (Prop. 84–86 / LADR 6.24, 6.25, 6.30)

For an orthonormal list: $\|\sum a_i e_i\|^2 = \sum |a_i|^2$, and the list is independent. For an orthonormal basis:

$$v = \sum_i \langle v, e_i\rangle e_i, \qquad \|v\|^2 = \sum_i |\langle v, e_i\rangle|^2.$$

The idea

Norm: Pythagoras handles two perpendicular vectors, and $a_m e_m$ is perpendicular to $a_1 e_1 + \cdots + a_{m-1} e_{m-1}$. So add one term at a time.

Independence: If $\sum a_i e_i = 0$, its norm $\sqrt{\sum |a_i|^2}$ is 0, so every $a_i = 0$.

Coordinates: To find a_k in $v = \sum a_i e_i$, take the inner product with e_k: the terms with $i \neq k$ give 0, and the k-th gives a_k. This is the reason to want orthonormal bases: coordinates without solving a linear system.

Proof. 1. $e_m \perp a_1 e_1 + \cdots + a_{m-1} e_{m-1}$, so Pythagoras adds $|a_m|^2$; induct.

2. $\sum a_i e_i = 0 \Rightarrow \sum |a_i|^2 = 0 \Rightarrow$ all $a_i = 0$.

3. If $v = \sum a_i e_i$, then $\langle v, e_k\rangle = \sum a_i \langle e_i, e_k\rangle = a_k$.

□

Parseval (infinite-dimensional version): for $f(x) = \sum c_n e^{2\pi i n x}$ on $[0,1]$, $\sum |c_n|^2 = \int_0^1 |f|^2$. With $f(x) = x = \frac{1}{2} - \frac{1}{2\pi i}\sum_{n\neq 0} \frac{1}{n} e^{2\pi i n x}$ this gives $\frac{1}{4} + \frac{1}{4\pi^2}\sum_{n\neq 0} \frac{1}{n^2} = \frac{1}{3}$, so $\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Theorem (Prop. 87 / LADR 6.32: Gram–Schmidt)

For independent $v_1, \ldots, v_n$: $e_1 = v_1/\|v_1\|$, and for $j \geq 2$

$$w_j = v_j - \sum_{k<j} \langle v_j, e_k\rangle e_k, \qquad e_j = w_j/\|w_j\|.$$

Then $e_1, \ldots, e_n$ is orthonormal and $\operatorname{span}(e_1, \ldots, e_j) = \operatorname{span}(v_1, \ldots, v_j)$ for every j.

The idea

To show: Orthonormal $e_1, \ldots, e_n$ with $\operatorname{span}(e_1, \ldots, e_j) = \operatorname{span}(v_1, \ldots, v_j)$ for every j.

For a unit vector e, $v - \langle v, e\rangle e \perp e$ (LADR 6.13 with $\|e\| = 1$). Subtracting a combination of $e_1, \ldots, e_{j-1}$ from v_j does not change the span of $e_1, \ldots, e_{j-1}, v_j$.

Build one vector at a time: from v_j subtract its component $\langle v_j, e_k\rangle e_k$ along each earlier e_k. Since the earlier e_k are perpendicular to each other, the subtraction for e_k does not change the inner products with the other e_i. Then divide by the length.

Only risk: $w_j = 0$, which would put v_j in $\operatorname{span}(v_1, \ldots, v_{j-1})$; independence rules it out.

Proof. 1. If $w_j = 0$, then $v_j \in \operatorname{span}(e_1, \ldots, e_{j-1}) = \operatorname{span}(v_1, \ldots, v_{j-1})$, contradicting independence.

2. For $k < j$: $\langle w_j, e_k\rangle = \langle v_j, e_k\rangle - \langle v_j, e_k\rangle = 0$. And $\|e_j\| = 1$.

3. $e_j \in \operatorname{span}(v_1, \ldots, v_j)$, so the spans are nested; both have dimension j, so they are equal.

□

- Every finite-dimensional inner product space has an orthonormal basis; every orthonormal list extends to one (extend to a basis, then Gram–Schmidt; the first vectors do not change) (LADR 6.35, 6.36).
- An orthonormal list of length $\dim V$ is an orthonormal basis (Prop. 85, T2).
- If the v_j are dependent, $w_j = 0$ exactly when $v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$: skip it and continue.
- **Schur (LADR 6.38):** over $\mathbb{C}$, T is upper-triangular in some *orthonormal* basis. Take a triangular basis (T7) and apply Gram–Schmidt: it keeps the spans $\operatorname{span}(v_1, \ldots, v_j)$, which are invariant, so the matrix stays triangular.

Example (Ex. 70: Gram–Schmidt on $\mathcal{P}_2$, $\langle p, q\rangle = \int_0^1 p\bar{q}$) 1. $\|1\| = 1$: $e_1 = 1$.

2. $w_2 = x - \langle x, 1\rangle = x - \frac{1}{2}$, $\|w_2\|^2 = \frac{1}{12}$: $e_2 = \sqrt{3}(2x-1)$.

3. $w_3 = x^2 - \frac{1}{3} - \frac{1}{2}(2x-1) = x^2 - x + \frac{1}{6}$, $\|w_3\| = \frac{1}{6\sqrt{5}}$: $e_3 = \sqrt{5}(6x^2 - 6x + 1)$.

Order of work: w_j, then $\|w_j\|^2$, then divide. Check $w_j \perp$ the earlier e's. (These are the shifted Legendre polynomials; on $[-1, 1]$, HW10 P1, the Legendre polynomials.)

Proposition (Prop. 88 / LADR 6.42, 6.58: Riesz representation)
For finite-dimensional V, every $\psi \in V'$ is $\psi(u) = \langle u, v\rangle$ for a unique v. In an orthonormal basis, $v = \sum_j \overline{\psi(e_j)} e_j$.

The idea

To show: A vector v with $\psi(u) = \langle u, v\rangle$ for all u.

On $\mathbb{R}^n$ a functional is $\psi(u) = a \cdot u$ for a row vector a, so $v = a$. A linear functional is determined by its values on a basis.

Write the unknown $v = \sum a_j e_j$ in an orthonormal basis and test on e_i: $\langle e_i, v\rangle = \bar{a}_i$ must equal $\psi(e_i)$. So $a_i = \overline{\psi(e_i)}$, and this v works because both sides agree on a basis.

Proof. 1. With $v = \sum \overline{\psi(e_j)} e_j$: $\langle e_i, v\rangle = \psi(e_i)$. Two linear functionals equal on a basis are equal.

2. Unique: if $\langle u, v\rangle = \langle u, v'\rangle$ for all u, take $u = v - v'$ to get $\|v - v'\|^2 = 0$.

(The notes use $\Gamma(v) = \langle -, v\rangle$: real-linear, not complex-linear since $\Gamma(iv) = -i\Gamma(v)$, injective, and $\dim_{\mathbb{R}} V = \dim_{\mathbb{R}} V'$.) □

Definition (Def. 47 / LADR 6.46, 6.47)**.** $U^\perp = \{v : \langle v, u\rangle = 0 \text{ for all } u \in U\}$, a subspace (linearity in the first slot).

- $\{0\}^\perp = V$; $V^\perp = \{0\}$; $U \cap U^\perp = \{0\}$ (a vector perpendicular to itself is 0).
- $U \subseteq W \Rightarrow W^\perp \subseteq U^\perp$.
- $(U + W)^\perp = U^\perp \cap W^\perp$ (perpendicular to all $u + w$ iff to all u and all w).
- $(U \cap W)^\perp = U^\perp + W^\perp$ (finite dim., HW10 P4): take $\perp$ of $(U^\perp + W^\perp)^\perp = U \cap W$ (Prop. 90).
- Finite dim.: $U^\perp = \{0\} \iff U = V$; $U^\perp = V \iff U = \{0\}$ (Prop. 89).
- **Computing $U^\perp$:** require $\langle v, u_i\rangle = 0$ for a spanning list of U and solve the linear system. Under Riesz, $U^\perp$ matches the annihilator U^0.

Theorem (Prop. 89 / LADR 6.49, 6.50)
If U is finite-dimensional, $V = U \oplus U^\perp$. If V is finite-dimensional, $\dim U^\perp = \dim V - \dim U$.

The idea

To show: Every v is $u + w$ with $u \in U$ and $w \perp U$, in exactly one way.

For a line $U = \text{span}(e)$, $\|e\| = 1$, this is LADR 6.13: $u = \langle v, e\rangle e$.

For a subspace, take an orthonormal basis $e_1, \ldots, e_m$ of U and use all the components: $u = \sum \langle v, e_k\rangle e_k$. Then $\langle v - u, e_k\rangle = 0$ for every k, so $v - u \perp U$. Uniqueness: $U \cap U^\perp = 0$, since a vector perpendicular to itself is 0.

Dimension: $\dim U^\perp = \dim V - \dim U$ then follows from the direct sum.

Proof. 1. Direct: if $u \in U \cap U^\perp$, then $\langle u, u\rangle = 0$, so $u = 0$.

2. Take an orthonormal basis $e_1, \ldots, e_m$ of U (Gram–Schmidt). Set $u = \sum \langle v, e_k\rangle e_k \in U$, $w = v - u$.

3. $\langle w, e_k\rangle = \langle v, e_k\rangle - \langle v, e_k\rangle = 0$ for each k, so $w \perp U$, and $v = u + w$.

□

Proposition (Prop. 90 / LADR 6.52)
If U is finite-dimensional, $(U^\perp)^\perp = U$.

The idea

To show: $(U^\perp)^\perp = U$. The inclusion $U \subseteq (U^\perp)^\perp$ is immediate from the definitions; the work is showing that $(U^\perp)^\perp$ is not bigger.

Finite dimension: Count: $\dim(U^\perp)^\perp = n - (n - \dim U) = \dim U$ (Prop. 89 twice), and a subspace of the same dimension is equal (T2).

Take $v \in (U^\perp)^\perp$ and split $v = u + w$ (Prop. 89). Then $w \in U^\perp$, and also $w = v - u \in (U^\perp)^\perp$. So w is perpendicular to itself, and $w = 0$.

Proof. $U \subseteq (U^\perp)^\perp$: every $u \in U$ is perpendicular to every element of $U^\perp$. Conversely let $v \in (U^\perp)^\perp$, $v = u + w$ ($u \in U$, $w \in U^\perp$). Then $w = v - u \in (U^\perp)^\perp$ and $w \in U^\perp$, so $\langle w, w\rangle = 0$ and $w = 0$. □

Definition (Def. 48 / LADR 6.55)**.** For finite-dimensional U, the **orthogonal projection** is $P_U(u + w) = u$ ($u \in U$, $w \in U^\perp$). It is the unique projector with range $P = U$, null $P = U^\perp$.

- **Uniqueness:** if $P^2 = P$, range $P = U$, null $P = U^\perp$, then $u = Px$ gives $Pu = P^2x = u$, so $P(u + w) = u$.
- **Properties** (LADR 6.57), all from $v = P_U v + (v - P_U v)$:
  - P_U is linear; $P_U^2 = P_U$; range $P_U = U$; null $P_U = U^\perp$; $v - P_U v \in U^\perp$.
  - $\|P_U v\| \le \|v\|$ (Pythagoras).
  - $I - P_U = P_{U^\perp}$ if V is finite-dimensional ($V = U^\perp \oplus U$).
- **Any projector** ($P^2 = P$) has $V = \text{null } P \oplus \text{range } P$ (Prop. 91). Idea: Pv is the obvious range part, and $P(v - Pv) = Pv - P^2v = 0$. If $w = Pv \in \text{null } P$, then $w = P^2v = Pw = 0$.
- It is orthogonal only when null $P \perp$ range P: $P(x, y) = (0, y)$ is $P_{\text{span}(e_2)}$; $P(x, y) = (x + y, 0)$ is not orthogonal.

Proposition (Prop. 92 / LADR 6.57(i))
If $e_1, \dots, e_m$ is an orthonormal basis of U: $P_U v = \sum_k \langle v, e_k\rangle e_k$.

The idea

To show: The coordinates of $P_U v$ in an orthonormal basis $e_1, \dots, e_m$ of U.

Coordinates in an orthonormal basis are inner products: $P_U v = \sum \langle P_U v, e_k\rangle e_k$ (Prop. 86 in U).

$v - P_U v \perp e_k$, so $\langle v, e_k\rangle = \langle P_U v, e_k\rangle$. The coordinates can be computed from v, without knowing $P_U v$ first.

Proof. $v = u + w$: $\langle v, e_k\rangle = \langle u, e_k\rangle$, so $P_U v = u = \sum \langle u, e_k\rangle e_k = \sum \langle v, e_k\rangle e_k$ (Prop. 86 in U). □

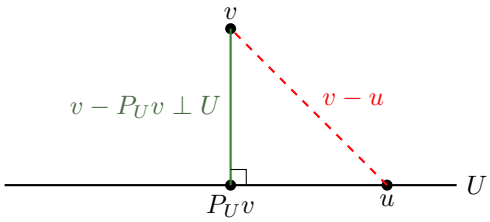
Proposition (HW10 P5, LADR 6.61: best approximation)
For $u \in U$: $\|v - P_U v\| \le \|v - u\|$, equality only at $u = P_U v$. So the distance from v to U is $\|v - P_U v\|$.

The idea

To show: Among $u \in U$, $\|v - u\|$ is smallest at $u = P_U v$.

Split the error: $v - u = (v - P_U v) + (P_U v - u)$. The first part is in $U^\perp$ and does not depend on u; the second is in U. They are perpendicular, so $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2$, which is smallest exactly when the second term is 0.

Proof. $v - u = (v - P_U v) + (P_U v - u)$ with pieces in $U^\perp$ and U, so $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2$. □



$v - u$ is the hypotenuse of a right triangle. One leg, $v - P_U v$, does not depend on u; the other, $P_U v - u$, lies in U and is 0 only when $u = P_U v$.

- **Bessel:** for any orthonormal $e_1, \dots, e_m$, $\sum |\langle v, e_k \rangle|^2 \le \|v\|^2$. The left side is $\|P_U v\|^2$ with $U = \operatorname{span}(e_k)$.
- **HW10 P7:** a projector is orthogonal iff $\|Pv\| \le \|v\|$ for all v. Idea for $\Leftarrow$: suppose $u \in \operatorname{range} P$ and $w \in \operatorname{null} P$ are not perpendicular. The point of the line $u + tw$ closest to 0 has $t = -\langle u, w \rangle / \|w\|^2 \neq 0$ and is shorter than u. But $P(u + tw) = u$, so $\|P(u + tw)\| > \|u + tw\|$. (Details in the solutions.)
- **Computing $P_U v$:** (1) Gram–Schmidt on U, then Prop. 92. (2) Or write $P_U v = \sum c_i u_i$ for a basis u_i and solve $\langle v - \sum c_i u_i, u_j \rangle = 0$ for all j. Check that $v - P_U v \perp$ each u_j.

Example (Ex. 71)

$P_U(x^3)$, $U = \operatorname{span}(1, x)$, $\langle p, q \rangle = \int_0^1 pq$. With $e_1 = 1$, $e_2 = \sqrt{3}(2x - 1)$: $\langle x^3, e_1 \rangle = \frac{1}{4}$, $\langle x^3, e_2 \rangle = \frac{3\sqrt{3}}{20}$, so $P_U(x^3) = \frac{1}{4} + \frac{9}{20}(2x - 1) = \frac{9}{10}x - \frac{1}{5}$. Check: $x^3 - \frac{9}{10}x + \frac{1}{5}$ is perpendicular to 1 and x.

Traps

- Orthogonal vectors are independent only if nonzero.
- $P^2 = P$ does not make P orthogonal: $P(x, y) = (x + y, 0)$ has $\|P(1, 1)\| = 2 > \sqrt{2}$.
- $V = U \oplus U^\perp$ can fail for infinite-dimensional U: in ℓ^2, the finitely supported sequences have $U^\perp = \{0\}$. Also ℓ^2 has no orthonormal (Hamel) basis.
- Gram–Schmidt depends on the order and on the inner product; normalize only after subtracting.
- Riesz over $\mathbb{C}$: the coefficients are conjugated.

§3 Problem set

HW10 P1. On $\mathcal{P}_2(\mathbb{R})$ use the inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$. Apply the Gram–Schmidt procedure to the ordered basis $1, x, x^2$ to obtain an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

HW10 P2. Let $W = \operatorname{span}\big((1,-1,1,0),(0,2,1,0)\big) \subseteq \mathbb{R}^4$ with the standard inner product. Find a basis of $W^{\perp}$.

HW10 P3. On $\mathcal{P}_2(\mathbb{R})$ use the same inner product as in Problem 1, and let $W = \operatorname{span}(x+1)$. Find a basis of $W^{\perp}$.

HW10 P4. Let U and W be subspaces of a finite-dimensional inner-product space V. Prove $(U \cap W)^{\perp} = U^{\perp} + W^{\perp}$.

HW10 P5. Let U be a subspace of a finite-dimensional inner-product space V, and let $e_1, \dots, e_m$ be an orthonormal basis of U.

(a) Prove that $P_U v = \sum_{j=1}^{m} \langle v, e_j\rangle e_j$.

(b) Prove the best-approximation property $\|v - P_U v\| \le \|v - u\|$ for every $u \in U$, and determine when equality holds.

HW10 P6. (optional) On $\mathcal{P}_1(\mathbb{R})$ use the inner product $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$. Find the unique polynomial $p \in \mathcal{P}_1(\mathbb{R})$ such that $\int_0^2 f(x)\,dx = \langle f, p\rangle$ for every $f \in \mathcal{P}_1(\mathbb{R})$.

HW10 P7. (optional) Let P be a projector on a finite-dimensional inner-product space, so $P^2 = P$. Prove that P is an orthogonal projection if and only if $\|Pv\| \le \|v\|$ for every v.

O1. Find the polynomial $p \in \mathcal{P}_1(\mathbb{R})$ that minimizes $\int_0^1 \big(x^2 - p(x)\big)^2 dx$, and the minimum value.

O2. In $\mathbb{R}^4$ with the dot product let $U = \operatorname{span}\big((1,1,1,1)\big)$ and $v = (1,2,3,4)$. Find $P_U v$, a basis of $U^{\perp}$, and the distance from v to U.

O3. Reflections. Let U be a subspace of a finite-dimensional inner product space and $R = I - 2P_U$. Prove that $\|Rv\| = \|v\|$ for all v and that $R^2 = I$.